Infinite matching theory may seem rather mature and complete as it stands, but there are still fascinating unsolved problems in the Erdős-Menger spirit concerning related discrete structures, such as partially ordered sets or hypergraphs. We conclude with one about graphs.

strongly perfect

weakly perfect

Call an infinite graph G *perfect* if every induced subgraph $H \subseteq G$ has a complete subgraph K of order $\chi(H)$, and *strongly perfect* if K can always be chosen so that it meets every colour class of some $\chi(H)$-colouring of H. (Exercise 75 gives an example of a perfect graph that is not strongly perfect.) Call G *weakly perfect* if the chromatic number of every induced subgraph $H \subseteq G$ is at most the supremum of the orders of its complete subgraphs.

Conjecture. (Aharoni & Korman 1993)
Every weakly perfect graph without infinite independent sets of vertices is strongly perfect.

8.5 Recursive structures

In this section we introduce another tool that is commonly used in infinite graph theory: to define a class of graphs recursively, so as to be able later to prove assertions about these graphs by (transfinite) induction. Rather than attempting a systematic treatment of this technique we give two examples; more can be found in the exercises.

Our first example is very simple: it describes the structure of a tree by recursively pruning away leaves and isolated ends. Let T be any tree, equipped with a root and the corresponding tree-order on its vertices. We recursively label the vertices of T by ordinals, as follows. Given an ordinal α, assume that we have decided for every $\beta < \alpha$ which of the vertices of T to label β, and let T_α be the subgraph of T induced by the vertices that are still unlabelled. Assign label α to every vertex t of T_α whose up-closure $\lfloor t \rfloor_{T_\alpha} = \lfloor t \rfloor_T \cap T_\alpha$ in T_α is a chain. The recursion terminates at the first α not used to label any vertex; for this α we put

T^*

$T_\alpha =: T^*$.

For each α, the vertices labelled α form an up-set in T_α: if $\lfloor t \rfloor_{T_\alpha}$ is a chain, then so is $\lfloor t' \rfloor_{T_\alpha}$ for every $t' \in \lfloor t \rfloor_{T_\alpha}$. Every T_α, therefore, is a down-set in T (induction on α) and hence connected. Thus, T_α is a tree, and the set of vertices labelled α induces in T_α a disjoint union of paths.

recursively prunable

Let us call T *recursively prunable* if every vertex of T gets labelled in this way, i.e., if $T^* = \emptyset$. We may then be able to prove assertions about T, or about graphs containing T as a normal spanning tree, by dealing in turn with those chains as they get deleted. The following proposition shows that the recursively prunable trees form a natural class also in structural terms:

Proposition 8.5.1. *A rooted tree is recursively prunable if and only if it contains no subdivision of the infinite binary tree T_2 as a subgraph.*

Proof. Let T be any rooted tree. Suppose first that T is not recursively prunable, i.e. that $T^* \neq \emptyset$. Since no vertex of T^* gets labelled when the recursion terminates, every $t \in T^*$ has two incomparable vertices of T^* above it. As T^* is connected, it is now easy to find a subdivision of T_2 in T^* inductively, along the levels of T_2.

Conversely, suppose that T contains a subdivision T' of T_2. We shall see in a moment that T' can be chosen 'upwards' in T, that is, in such a way that the tree-order which T induces on its vertices agrees with its own tree-order as induced by T_2. If this is the case, then every vertex of T' has two incomparable vertices of T' above it (in both orders). Hence there can be no minimal ordinal α such that a vertex of T' is labelled α. Thus all of T' remains unlabelled, and $\emptyset \neq T' \subseteq T^*$ as desired.

It remains to show that T' can indeed be chosen in this way. Let T' be any subdivision of T_2 in T, and let u be minimal in the tree-order of T among the vertices of T'. Induction on the levels of the tree $\lfloor u \rfloor_{T'}$ shows that $\leqslant_{T'}$ and $\leqslant_T$ agree on $\lfloor u \rfloor_{T'}$: any upper neighbour in T' of a vertex $t \in \lfloor u \rfloor_{T'}$ must lie above t also in T, since the unique lower neighbour of t in T is either not in T' (if $t = u$), or by induction it is the unique lower neighbour of t also in T'. Pick any branch vertex v of T' in $\lfloor u \rfloor_{T'}$. Then $\lfloor v \rfloor_{T'}$ is the desired subdivision of T_2 in T. □

The charm of the recursive pruning discussed above lies in the fact that it removes the 'messy bits' of a given tree in an automated sort of way: we do not have to know where they are, but if our given tree contains a 'clean' ever-branching subtree, then the recursion will reveal it.

And there is another way of viewing it. We might think of rooted paths (paths with a first vertex, which we take to be the root) as particularly basic objects, and call them *rooted trees of rank* 0. We could then define rooted trees of higher ordinal rank inductively, taking as the rooted trees of rank α those that do not have any rank $\beta < \alpha$ but in which it is possible to delete a path starting at the root so as to leave components that each have some rank $< \alpha$ when taken with the induced tree-order. Then the rooted trees that are assigned a rank in this way are precisely the recursively prunable ones, and those of rank $\leqslant \alpha$ are precisely those whose labels do not exceed α (Exercise 77).

We now apply the same idea to graphs that are not necessarily trees. Let us assign *rank* 0 to all the finite graphs. Given an ordinal $\alpha > 0$, we assign *rank* α to every graph G that does not already have a rank $\beta < \alpha$ and which has a finite set U of vertices such that every component of $G - U$ has some rank $< \alpha$. *rank*

When disjoint graphs G_i have ranks $\alpha_i < \alpha$, their union clearly has a rank of at most α; if the union is finite, it has rank $\max_i \alpha_i$. Induction on α shows that subgraphs of graphs of rank α also have a rank of at most α. Conversely, joining finitely many new vertices to a graph, no matter how, will not change its rank.

Not every graph has a rank. Indeed, the ray cannot have a rank, since deleting finitely many of its vertices always leaves a component that is also a ray. As subgraphs of graphs with a rank also have a rank, this means that only rayless graphs can have a rank. But all these do:

Lemma 8.5.2. *A graph has a rank if and only if it is rayless.*

Proof. Consider a graph G that has no rank. Then one of its components, C_0 say, has no rank; let v_0 be a vertex in C_0. Now $C_0 - v_0$ has a component C_1 that has no rank; let v_1 be a neighbour of v_0 in C_1. Continuing inductively, we find a ray $v_0 v_1 \dots$ in G. □

Because of Lemma 8.5.2, we call the ranking defined above the *ranking of rayless graphs*. As an application of this ranking, we now prove the unfriendly partition conjecture from Section 8.1 for rayless graphs.

Theorem 8.5.3. *Every countable rayless graph G has an unfriendly partition.*

Proof. To help with our formal notation, we shall think of a partition of a set V as a map $\pi \colon V \to \{0,1\}$. We apply induction on the rank of G. When this is zero then G is finite, and an unfriendly partition can be obtained by maximizing the number of edges across the partition.
α Suppose now that G has rank $\alpha > 0$, and assume the theorem as true for graphs of smaller rank.

U Let U be a finite set of vertices in G such that each of the compo-
$C_0, C_1, \dots$ nents $C_0, C_1, \dots$ of $G - U$ has rank $< \alpha$. Partition U into the set U_0 of vertices that have finite degree in G, the set U_1 of vertices that have
U_0, U_1, U_2 infinitely many neighbours in some C_n, and the set U_2 of vertices that have infinite degree but only finitely many neighbours in each C_n.

$G_0, G_1, \dots$ For every $n \in \mathbb{N}$ let $G_n := G[U \cup V(C_0) \cup \dots \cup V(C_n)]$. This is a graph of some rank $\alpha_n < \alpha$, so by induction it has an unfriendly partition π_n. Each of these π_n induces a partition of U. Let π_U be a partition of U induced by π_n for infinitely many n, say for $n_0 < n_1 < \dots$. Choose n_0 large enough that G_{n_0} contains all the neighbours of vertices in U_0, and the other n_i large enough that every vertex in U_2 has more
$n_0, n_1, \dots$ neighbours in $G_{n_i} - G_{n_{i-1}}$ than in $G_{n_{i-1}}$, for all $i > 0$. Let π be the partition of G defined by letting $\pi(v) := \pi_{n_i}(v)$ for all $v \in G_{n_i} - G_{n_{i-1}}$
π and all i, where $G_{n_{-1}} := \emptyset$. Note that $\pi|_U = \pi_{n_0}|_U = \pi_U$.

Let us show that π is unfriendly. We have to check that every vertex is *happy with* π, i.e., that it has at least as many neighbours in the opposite class under π as in its own.[8] To see that a vertex $v \in G-U$ is happy with π, let i be minimal such that $v \in G_{n_i}$ and recall that v was happy with π_{n_i}. As both v and its neighbours in G lie in $U \cup V(G_{n_i} - G_{n_{i-1}})$, and π agrees with π_{n_i} on this set, v is happy also with π. Vertices in U_0 are happy with π, because they were happy with π_{n_0}, and π agrees with π_{n_0} on U_0 and all its neighbours. Vertices in U_1 are also happy. Indeed, every $u \in U_1$ has infinitely many neighbours in some C_n, and hence in some G_{n_i}; let i be minimal with this property. Then u has infinitely many opposite neighbours under π_{n_i} in G_{n_i}, and hence in $G_{n_i} - G_{n_{i-1}}$. Since π_{n_i} agrees with π on both U and $G_{n_i} - G_{n_{i-1}}$, our vertex u has infinitely many opposite neighbours also under π. Vertices in U_2, finally, are happy with every π_{n_i}. By our choice of n_i, at least one of their opposite neighbours under π_{n_i} must lie in $G_{n_i} - G_{n_{i-1}}$. Since π_{n_i} agrees with π on both U_2 and $G_{n_i} - G_{n_{i-1}}$, this gives every $u \in U_2$ at least one opposite neighbour under π in every $G_{n_i} - G_{n_{i-1}}$. Hence u has infinitely many opposite neighbours under π, which clearly makes it happy. $\square$

8.6 Graphs with ends: the complete picture

In this section we shall develop a deeper understanding of the global structure of infinite graphs, especially locally finite ones, that can be attained only by studying their ends. This structure is intrinsically topological, because topology best captures our intuition about convergence.[9]

Our first goal will be to make precise our intuitive idea that the ends of a graph are the 'points at infinity' to which its rays converge. To do so, we shall define a topological space $|G|$ associated with a graph $G = (V, E, \Omega)$ and its ends.[10] By considering topological versions of paths, cycles and spanning trees in this space, we shall then be able to extend to infinite graphs some parts of finite graph theory that would not otherwise have infinite counterparts; see the notes for more examples. Thus, the ends of an infinite graph turn out to be more than a curious phenomenon: they form an integral part of the picture, without which it cannot be properly understood. V, E, Ω

To build the space $|G|$ formally, we start with the set $V \cup \Omega$. For every edge $e = uv$ we add a set $\mathring{e} = (u, v)$ of continuum many points, making these sets $\mathring{e}$ disjoint from each other and from $V \cup \Omega$. We then choose for each e some fixed bijection between $\mathring{e}$ and the real interval $(0, 1)$, and (u, v)

[8] It is only by tradition that such partitions are called 'unfriendly'; our vertices love them.

[9] Only point-set topology is needed for the text. See the exercises for more.

[10] The notation of $|G|$ comes from topology and clashes with our notation for the order of G. But there is little danger of confusion, so we keep both.